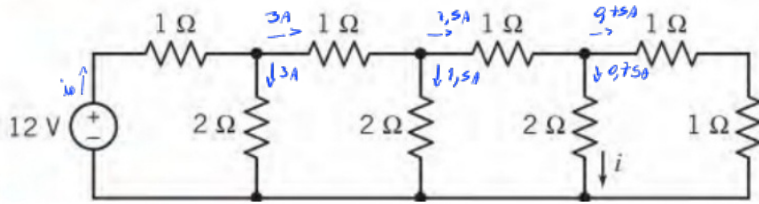


P 3.6-3 Determine o valor de i no circuito da Figura P 3.6-3 usando as reduções necessárias e o princípio de divisão de corrente.

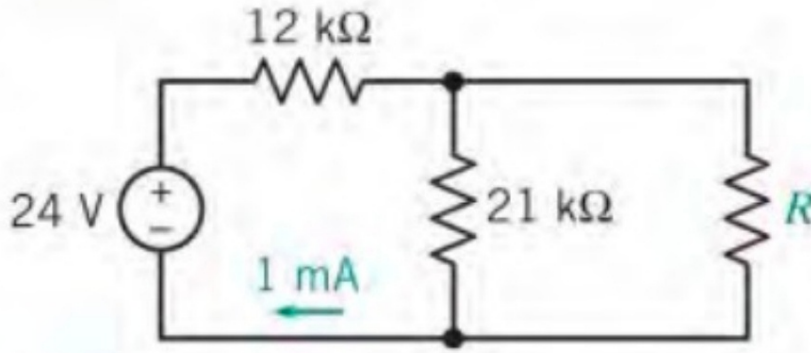


$$R_{\text{equivalente}} = 2\Omega$$

$$12 = 2 \cdot i$$

$$i_0 = 6A$$

* Como os resistores são equivalentes para os pontos de divisão por igual.



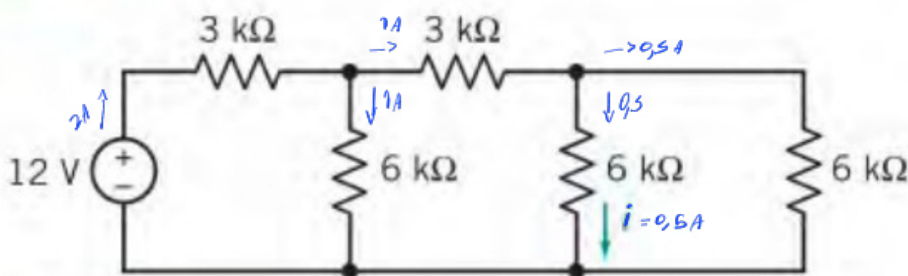
$$R_{eq} = \frac{R \cdot 21 \cdot 10^3}{R + 21 \cdot 10^3} + 12 \cdot 10^3 = 10^3 \left(\frac{R \cdot 21}{R + 21} + 12 \right)$$

$$= 10^3 \left(\frac{21R + 12R + 252}{R + 21} \right) = 10^3 \left(\frac{33R + 252}{R + 21} \right)$$

$$* V = R \cdot i$$

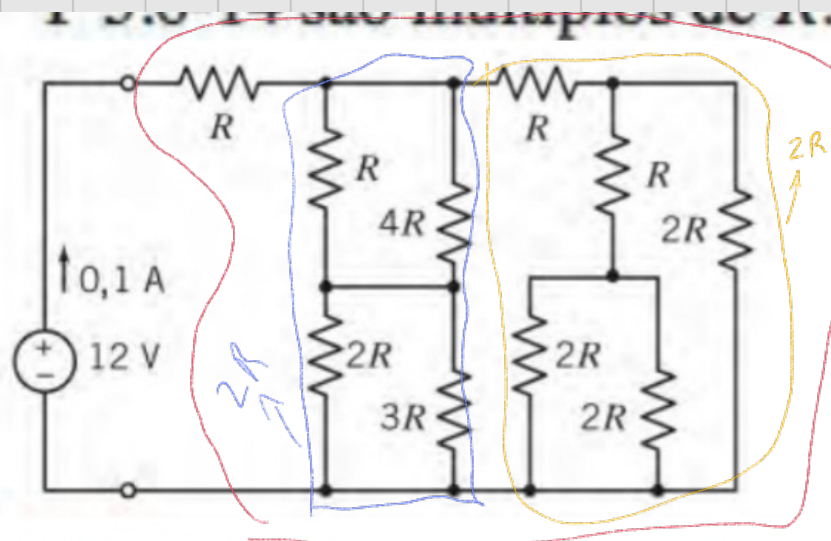
$$24 = 10^3 \left(\frac{33R + 252}{R + 21} \right) \cdot 10^{-3}$$

$$24R + 504 = 33R + 252 \quad \therefore R = 28\Omega$$



$$12 = 6 \cdot i_0$$

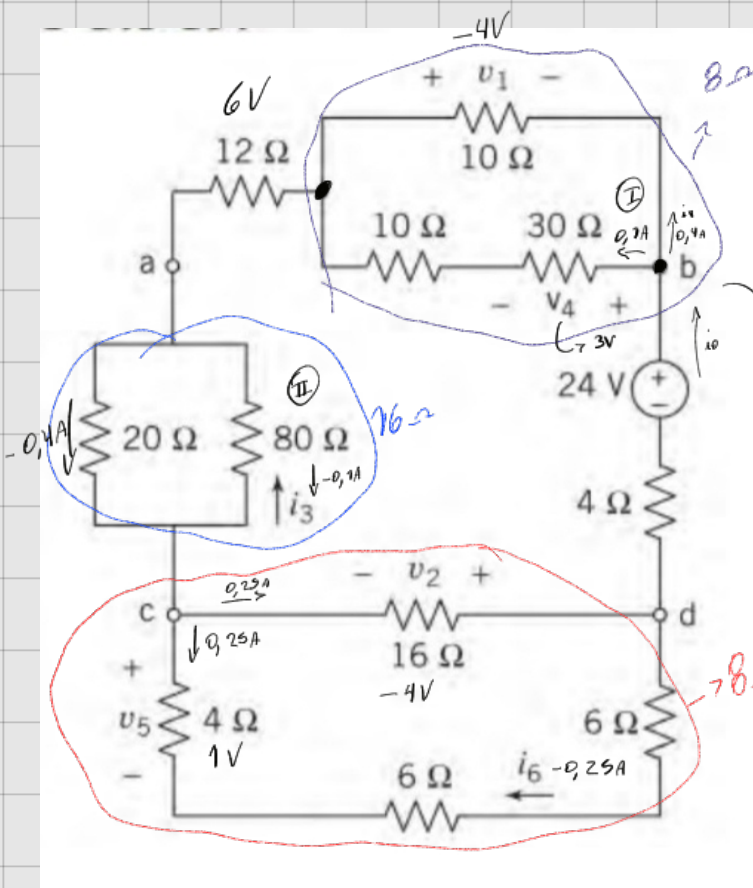
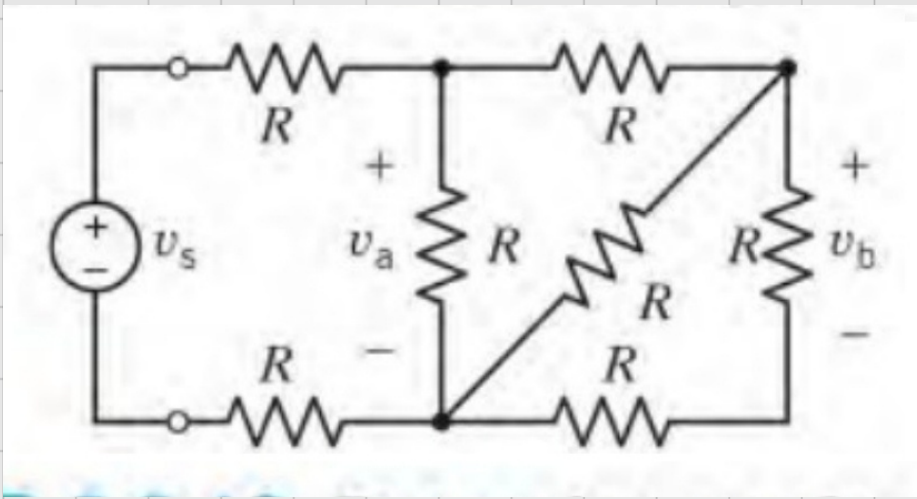
$$i_0 = 2A$$



$$12 = 2R \cdot 0,1$$

$$120 = 2R$$

$$R = 60\Omega$$



Resistencia: 48Ω

$$24 = 48 \cdot i_o$$

$$V = R \cdot i \rightarrow i_o = 0,5$$

$V_1 = -4V$	$i_3 = -0,1A$
$V_2 = -4V$	$i_6 = -0,25A$
$V_3 = 3V$	
$V_5 = 1V$	

* Unión de corriente:

$$i_{corriente} = i_{total} \left(\frac{R_{opuesta}}{R_{total\ de\ paralelo}} \right)$$

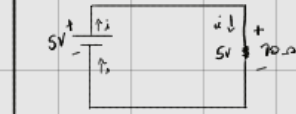
$$I \quad i_4 = 0,5 \left(\frac{10}{8} \right) = 0,4$$

$$II \quad i_5 = 0,5 \left(\frac{29}{100} \right) = 0,1$$

$$I_1 = 0,5 \left(\frac{8}{10} \right)$$

$$> 0,5 \cdot \frac{10}{50} = 0,1 \text{ p/línea} \quad 0,5 \cdot \frac{40}{50} = 0,4 \text{ p/línea.}$$

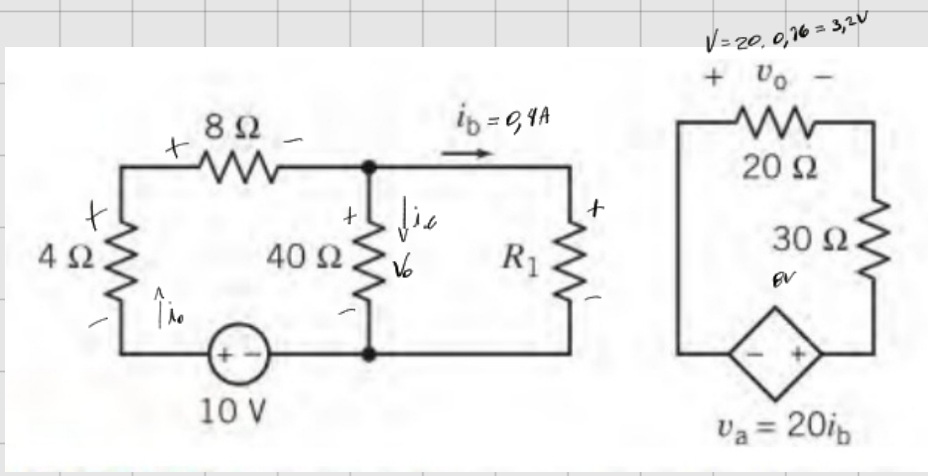
* Corriente siempre va de + para -



sentido de las corrientes

* Alínea: - p/+

* Paralela: + p/-



* Sabemos que $V_o = 8V$

$$\therefore 8 = 20 i_b \rightarrow b = 0,4A$$

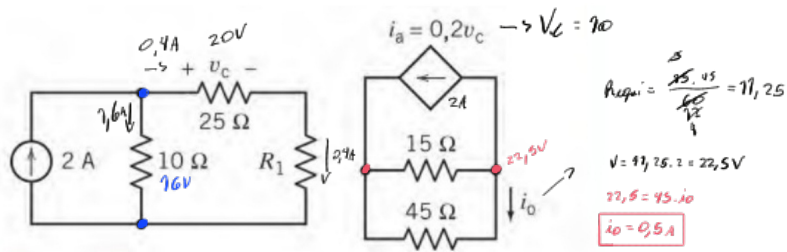
$$\frac{V_o}{40} + 0,4 = I$$

$$10 - (12 \cdot I) = V_o$$

$$10 - \left[12 \cdot \left(\frac{V_o}{40} + 0,4 \right) \right] = V_o$$

$$V_o = 4V \quad \therefore R = 20\Omega$$

P 3.6-41 Considere o circuito da Figura P 3.6-41. Dado que a corrente da fonte de corrente dependente é $i_a = 2$ A, determine o valor de R_1 e i_o .



Handwritten calculations for P 3.6-41:

$$V = R \cdot i$$

$$10 = 25 i_1$$

$$i_1 = \frac{10}{25} = 0.4$$

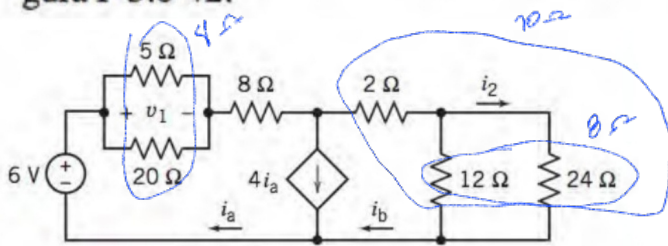
$$10 = v_c + v_{R1}$$

$$10 = 10 \cdot V_{R1}$$

$$V_{R1} = 0 \therefore 0 = R_1 \cdot 0.4$$

$$R_1 = 15\Omega$$

P 3.6-42 Determine o valor de i_a , i_b , i_2 e v_1 no circuito da Figura P 3.6-42.



Lei das Correntes:

$$i_o = 4i_a + i_b$$

$$i_b = -3i_o$$

Handwritten equations for P 3.6-42:

$$V_1 = 4 \cdot i_a$$

$$V_2 = 8 \cdot i_a$$

$$V_3 = 2 \cdot i_b = -6 \cdot i_a$$

$$V_4 = 8 \cdot i_b = -24 \cdot i_a$$

Handwritten equations for P 3.6-42:

$$V_1 + V_2 + V_3 + V_4 - 6 = 0$$

$$V_2 + V_3 + V_4 = 6$$

Handwritten equations for P 3.6-42:

$$-18i_o = 6$$

$$i_o = -\frac{2}{3}A \quad i_b = 1A$$